

SHOW OF HANDS...

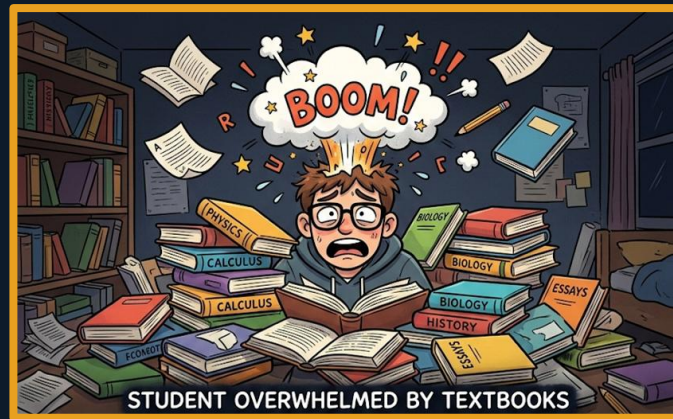
🧑 Who's pulled an all-nighter before a big exam?

🧑 Who read SO much your mind went blank mid-question?

🧑 Who wrote 3 great paragraphs — for the wrong question?

Your AI agent is doing exactly this.
Every. Single. Day.

The model isn't broken. The context is.





NEXTGEN DATA 2026

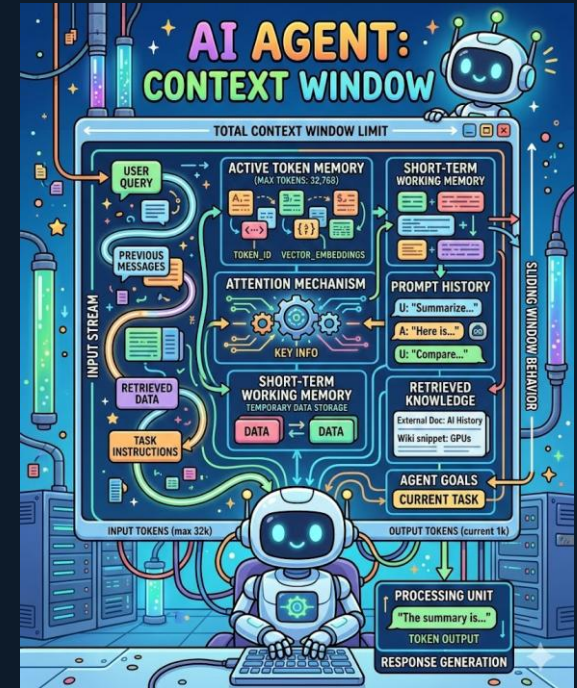
CONTEXT ENGINEERING

The Hidden Skill Behind Production-Ready AI

Talk By

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AI Engineer · Best-Selling Author · AI & Data Science Mentor · Speaker

- **Microsoft** (2021 – Present)
 - **Applied Scientist II** (2025 – Present) *[M365 Copilot Extensibility]*
 - Software Engineer II (2024 – 2025) *[M365 Copilot Extensibility]*
 - Data & Applied Scientist II (2023 - 2024) *[Bing Autosuggest + Search History]*
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- **IIT Bombay** (2017 – 21)
 - Major in Mechanical Engineering (9.90 CPI)
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 - **Institute Rank 2** | Department Rank 1
- Author of “Beyond Code” – a **National Bestseller**
- Author of “Low-Pass Filter” – **AI Newsletter**
- **7 US Patents** | 2 Research Papers
- AI & Data Science **Mentor**
 - Empowered **20,000+ students & professionals** in AI & Data Science



Today's Journey

01



Production
Reality Check

02



Context
Failures

03



Context
Strategies

04



Cutting-Edge
Patterns

05



Open Source
Arsenal

The Demo Worked. Then You Deployed It.



DEMO

- ✓ Perfect retrieval every time
- ✓ Fast, snappy responses
- ✓ Coherent, on-topic output
- ✓ No edge cases in sight

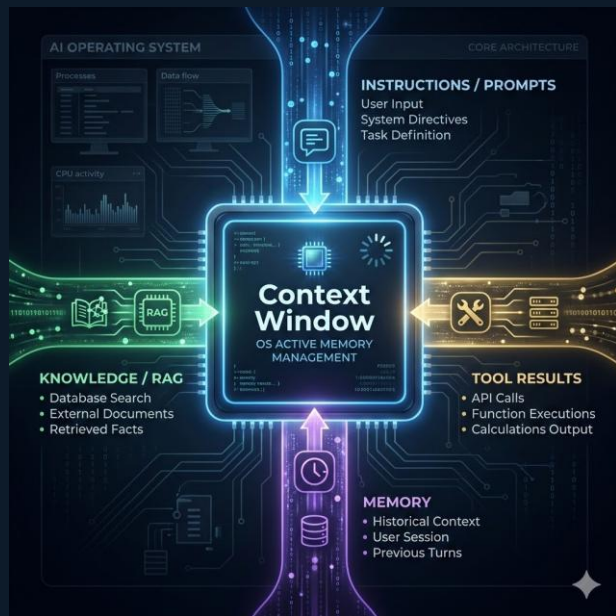


PRODUCTION

- ✗ Hallucinations everywhere
- ✗ Latency unbearable
- ✗ Costs spiralling out of control
- ✗ Edge cases are ALL the cases

It's NOT your model. It's your context.

The Context Window is Your RAM



Instructions

Prompts, system messages, few-shot examples, tool descriptions



Knowledge

Retrieved docs, facts, memories, RAG results



Tools

Tool call results, intermediate agent outputs

"...the delicate art and science of filling the context window with just the right information."

— Karpathy



The 4 Deadly Context Failures

"Every AI failure has a context fingerprint."

01

Context Distraction

02

Context Confusion

03

Context Clash

04

Context Saturation

Context Failures That Kill Your Agent

01 Context Distraction

"When too much history becomes a distraction"

Long contexts can make models over-focus on past tokens instead of reasoning

Smaller models hit the distraction ceiling earlier ($\approx 32k$ tokens in some studies))

02 Context Confusion

"When irrelevant context leads the model astray"

Adding too many tools or instructions forces the model to consider everything, even irrelevant options.

Example: Llama-3.1 8B failed with 46 tools but succeeded with only 19

03 Context Clash

"Contradictory instructions battling for control"

When system prompts, memory, and examples disagree, the model doesn't know which instruction wins.

Example: System says, "be concise", memory says "user likes detail", few-shot examples are long.

04 Context Saturation

"When the context window overflows..."

Long inputs can exceed the model's context window, leading to silent truncation of earlier tokens.

Example: Only the last 3 pages of a 10-page legal document are included, resulting in an incomplete summary

The 4 Strategies of Context Engineering

You can't put everything in. So, engineer what goes in...



WRITE

Save it before you lose it



SELECT

*The right piece,
not all pieces*



COMPRESS

*Meaning survives.
Tokens don't.*



ISOLATE

Separate what shouldn't mix



WRITE

Save it before you lose it

Scratchpads (within-session)

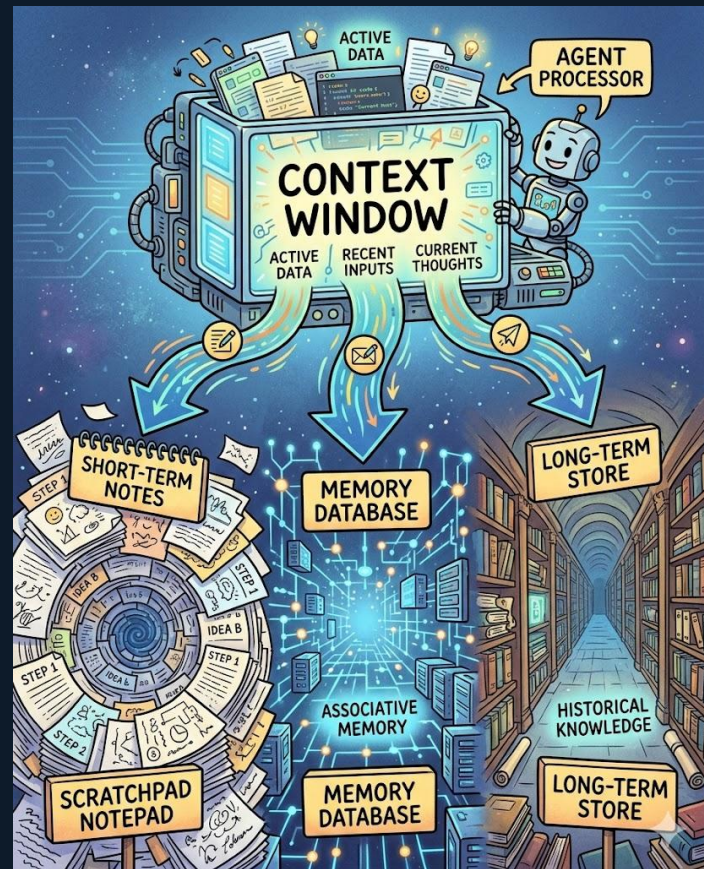
Agents externalise notes, plans & findings to survive context overflow

Anthropic's multi-agent researcher saves its plan BEFORE the 200K token limit is hit

Long-Term Memory (cross-session)

Reflexion, Generative Agents papers → self-reflection memories reused across runs

Used by: ChatGPT, Cursor, Windsurf for persistent user context



"If it's not written down, it doesn't exist."

SELECT

The right information, not all information



Memory Selection

Semantic (facts), **Episodic** (examples), **Procedural** (instructions) — each needs different retrieval strategies



Knowledge / RAG

Windsurf stack: AST parsing + embeddings + knowledge graphs + grep + re-ranking.

Embeddings alone break at scale.



Tool Selection

RAG on tool descriptions reduces overwhelm.

Recent papers: 3× better tool selection accuracy.

"Retrieval is not search. Retrieval is surgery."

COMPRESS

Meaning survives. Tokens don't have to.

Summarisation

Replace full history with rolling summaries

LangGraph's Summarization Node: trigger after N msgs

Preserve decisions & intent — discard raw dialogue

Trimming

Remove tool results no longer relevant

Claude's thinking tokens: useful for reasoning,
unnecessary in passed context

Keep structured state; strip verbose transcripts

THE ECONOMICS

WITHOUT COMPRESSION

50,000 tokens Full \$\$\$ · ↓ Lost in noise

WITH COMPRESSION

3,000 tokens -94% cost · ↑ Signal preserved

"A smaller context with the right information beats a bigger context with everything."



ISOLATE

Separate what shouldn't mix

Why it matters

Multi-agent systems: each subagent needs clean, focused context — not the full parent history

How leading systems do it

Anthropic's Multi-Agent Researcher: subagents get only their sub-problem

Enables parallelism

Isolated contexts → subagents run simultaneously → orchestrator synthesises results

The trade-off

More isolation = less contamination, more orchestration overhead. Worth it at scale.

"Build AI systems like microservices: each agent does one thing, with one clean context."

Advanced Tool Use — Where the Industry is Headed

The Problem: 50+ MCP tools in context = **55,000+ tokens** *before your agent reads a single request*



Tool Search Tool

Discover tools on-demand

Mark tools as deferred — loaded only when needed — 85% token reduction

Opus 4: 49% → 74% accuracy



Programmatic Tool Calling

Orchestrate via code

Write Python to call tools

Intermediate results stay OUT of context.

Only final output enters the window.

The Context Engineering Arsenal

FRAMEWORKS WITH BUILT-IN CONTEXT ENGINEERING BLOCKS



LangGraph

Isolated subgraph contexts ·
Summarization Node · Persistent
state



LangMem

Long-term memory · Semantic /
Episodic / Procedural memory types



LlamaIndex

Advanced RAG · Knowledge graphs ·
Context-aware chunking

DEDICATED CONTEXT MANAGEMENT

Mem0

Adaptive memory layer, cross-session persistence

Zep / Graphiti

Knowledge graph memory, temporal reasoning

Letta (MemGPT)

OS-inspired: core, archival & recall memory hierarchy

LangGraph Memory

Built-in memory store with thread-scoped & cross-session options

"You don't have to build context engineering from scratch. Start here."

That's All Folks!

For More Interesting Stuff on AI, Data Science & the Latest Technologies...



"Fill the context window with just the right information. Everything else is noise."